PMUS Trading HUD: Architecture & Metric Specification

Polymarket Global Market Making & Liquidity Rewards Real-Time Telemetry Guide

1. System Architecture & Overview

The PMUS Trading HUD is a real-time web application built on Streamlit that connects to the trading engine without imposing runtime overhead or process locking. The dashboard reads operational telemetry from SQLite storage running in Write-Ahead Logging (WAL) mode. This design allows the trading engine to write high-frequency order, fill, and liquidity reward snapshots concurrently while the user interface polls data every two seconds.

By utilizing reactive fragments (st.fragment), the user interface updates KPI cards, active order blotters, and market allocations in place without full page refreshes.

2. Top KPI Metric Cards

The top row of the dashboard presents five high-level performance indicators for rapid account assessment:

Metric Card	Underlying Source	Interpretation & Function
Balance / Collateral	Exchange API: client.get_balance_allowance()	Live USDC collateral held in your proxy wallet on Polygon. The sub-delta shows the configured maximum risk cap (PMUS_MAX_COLLATERAL). The bot halts new quote generation if total committed funds reach this threshold.
Accrued LIP Rewards	SQLite ledger: SUM(lip_snapshots.reward_usd)	Estimated maker rewards accrued during the active session. The daily pool rate is amortized per qualifying second based on your posted size relative to total qualifying pool depth.
Qualifying Seconds	SQLite ledger: qualifying_seconds counter	The exact number of seconds where posted quotes met Polymarket spread and size rules. The delta indicates the individual breakdown between qualifying buy-side seconds and sell-side seconds.
Total Fills & Volume	SQLite ledger: fills table count and sum	The number of executions where aggressive market participants crossed into resting orders. Volume represents the aggregate dollar value traded (Price * Quantity).
Realized P&L; & Rebates	SQLite ledger: Closed position matching	Net trading profit or loss realized when round-trip contract positions close. The sub-metric shows maker fee rebates credited by the exchange.

3. Market Quoting Overview Table

This table provides visibility into the bot's capital deployment across active liquidity pools. Each row represents a specific contract outcome:

Column Header	Value Representation	Trading Impact
Market Slug	256-bit token ID or asset identifier	Identifies the specific outcome token being quoted on the Polymarket CLOB.
Status	QUOTING vs MONITORING	QUOTING indicates resting limit orders exist in the book. MONITORING indicates the market is tracked by the scanner but capital is allocated elsewhere.
Qualifies	YES / NO flag	Indicates whether the current quotes meet both side requirements under the active max_spread threshold.
Bid Qual / Ask Qual	YES / NO side flags	Confirms whether the buy-side or sell-side orders independently satisfy the minimum size and maximum spread distance.
Best Bid / Bid Qty	Price ($) and contract volume	The bot's highest active buy price and posted contract size.
Best Ask / Ask Qty	Price ($) and contract volume	The bot's lowest active sell price and posted contract size.

Target Size	Minimum qualifying contract threshold	The minimum order size required by Polymarket rules to earn liquidity rewards (e.g. 50 or 200 contracts).
Est LIP Rate/s	Dollar reward rate per second	Calculated yield accrued each second while maintaining qualifying depth relative to competitor order sizes.

4. Fills Blotter & Open Positions

The Fills & Execution Blotter audits round-trip order executions in reverse chronological order. Columns include execution timestamp, contract slug, order side (BUY_LONG, SELL_LONG, BUY_SHORT, SELL_SHORT), execution price, contract count, fee impact (+ indicates maker rebate received, - indicates taker fee), and exchange order ID.

When fills accumulate into held inventory, the Open Positions section renders net contracts, volume-weighted average entry price, and total dollar cost committed to the position.

5. Critical Invariants & Rules

1. Minimum Payout Floor: Polymarket enforces that any market earning under $1.00 total across a 24-hour distribution window pays out zero. The market selector filters out candidates incapable of clearing this hurdle.

2. Shadow vs Live Execution: In shadow mode, the engine quotes against live exchange order books and simulates fills against real trade flow. In live mode, cryptographically signed EIP-712 orders are dispatched directly to the CLOB matching engine.

3. WAL Concurrency: SQLite operates in Write-Ahead Logging mode. You can inspect the dashboard or run CLI reports at any time without pausing the background trading daemon.